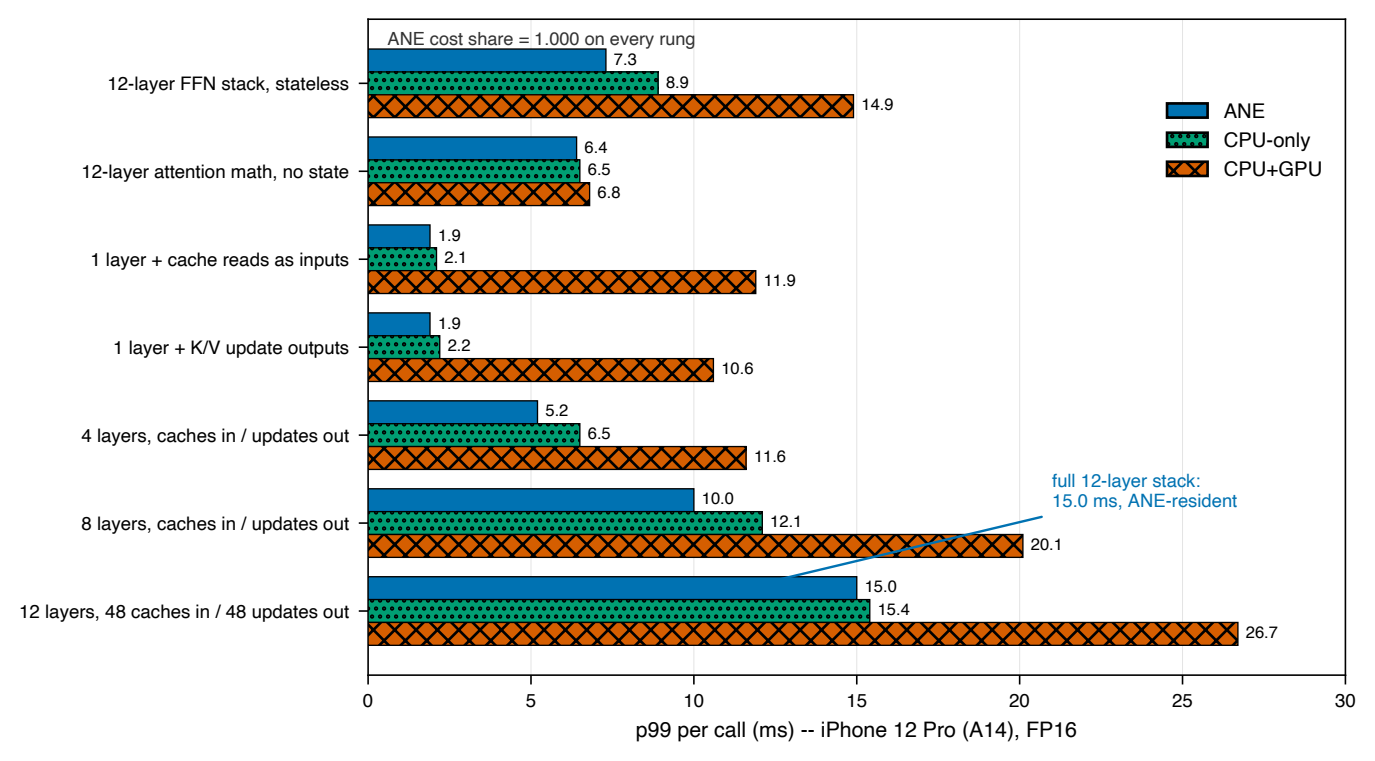

ANE cost share = 1.000 on every rung
12-layer FFN stack, stateless
7.3
8.9
14.9
12-layer attention math, no state
6.4
6.5
6.8
1 layer + cache reads as inputs
1.9
2.1
11.9
1 layer + K/V update outputs
1.9
2.2
10.6
4 layers, caches in / updates out
5.2
6.5
11.6
8 layers, caches in / updates out
10.0
12.1
20.1
12 layers, 48 caches in / 48 updates out
15.0
15.4
26.7
full 12-layer stack:
15.0 ms, ANE-resident
ANE
CPU-only
CPU+GPU
0
5
10
15
20
25
30
p99 per call (ms) -- iPhone 12 Pro (A14), FP16